

2.6 Limits Involving Infinity; Asymptotes of Graphs

In this section we investigate the behavior of a function when the magnitude of the independent variable x becomes increasingly large, or $x \rightarrow \pm\infty$. We further extend the concept of limit to infinite limits, which are not limits as before, but rather a new use of the term limit.

① Finite Limits as $x \rightarrow \pm\infty$

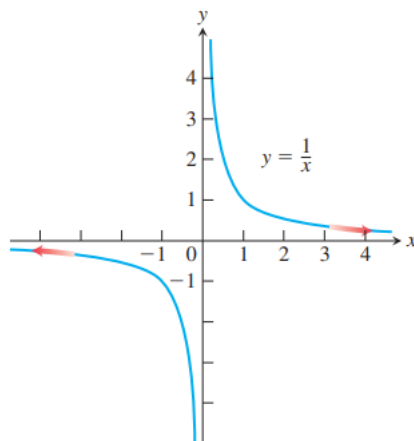
$\lim_{x \rightarrow \infty} f(x) = L \Leftrightarrow$ (a) As x gets bigger and bigger, $f(x)$ gets closer and closer to L .

(b) $\forall \varepsilon > 0, \exists M > 0$, such that $x > M \Rightarrow |f(x) - L| < \varepsilon$.

$\lim_{x \rightarrow -\infty} f(x) = L \Leftrightarrow$ (a) As x gets smaller and smaller, $f(x)$ gets closer and closer to L

(b) $\forall \varepsilon > 0, \exists M > 0$, such that $x < -M \Rightarrow |f(x) - L| < \varepsilon$.

For example, $f(x) = \frac{1}{x}$ has limit 0 as $x \rightarrow \infty$ or $x \rightarrow -\infty$, or that 0 is a limit of $f(x) = \frac{1}{x}$ at infinity and negative infinity.



Note 1 $\lim_{x \rightarrow \pm\infty} k = k$, k is a constant.

Note 2 $\lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$.

Note 3 All the limit laws in 2.2 are true for limits as $x \rightarrow \pm\infty$.

Example 1 Find the limits.

(a) $\lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right).$

(b) $\lim_{x \rightarrow \infty} \frac{\pi\sqrt{3}}{x^2}.$

② Limits at Infinity of Rational Functions

Recall For $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, $a_n \neq 0$, n is called the degree of $f(x)$, and $a_n x^n$ is called the leading term.

◆ To find the limit of a rational function $\frac{P(x)}{Q(x)}$ as $x \rightarrow \pm\infty$, we

(a) find the big degree of $P(x)$ and $Q(x)$, say t .

(b) divide both $P(x)$ and $Q(x)$ by x^t .

(c) use $\frac{1}{\infty} = 0$ to evaluate the result.

Example 2 Find the following limits.

(a) $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}.$

(b) $\lim_{x \rightarrow \infty} \frac{11x + 2}{2x^3 - 1}.$

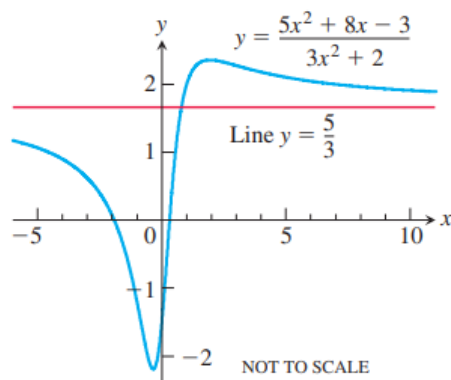
③ Limits at Infinity of other Functions

Example 3 Find $\lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 + 16} \right).$

④ Horizontal Asymptotes

DEFINITION A line $y = b$ is a horizontal asymptote of the graph of a function $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$.

For example, the graph of the function $f(x) = \frac{5x^2 + 8x - 3}{3x^2 + 2}$ in example 2 has the line $y = \frac{5}{3}$ as a horizontal asymptote on both the right and the left because $\lim_{x \rightarrow \infty} f(x) = \frac{5}{3}$ and $\lim_{x \rightarrow -\infty} f(x) = \frac{5}{3}$.



Example 4 Find the horizontal asymptotes of the graph of

$$f(x) = \frac{x^3 - 2}{|x|^3 + 1}.$$

Example 5 Using the sandwich theorem, find the horizontal asymptote of the curve

$$y = 2 + \frac{\sin x}{x}.$$

⑤ Infinite Limits

Let us look again at the function $f(x) = \frac{1}{x}$. As $x \rightarrow 0^+$, the values of f grow without bound, eventually reaching and surpassing every positive real number. That is, given any positive real number B , however large, the values of f become larger still.

Thus, f has no limit as $x \rightarrow 0^+$. It is nevertheless convenient to describe the behavior of f by saying that $f(x)$ approaches ∞ as $x \rightarrow 0^+$. We write

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty.$$

In writing this equation, we are not saying that the limit exists since there is no such a real number ∞ .

Note 4 $\frac{1}{0^+} = \infty$, $\frac{1}{0^-} = -\infty$.

Note 5 Some basic limits

(a) $\frac{1}{\infty} = 0$.

(b) $\frac{1}{0} = \infty$.

(c) $\frac{0}{0}$ ----- canceling the zero factor and trying again.

(d) $\frac{A}{B}$ ($B \neq 0$).

Example 6 Find the following limits.

(a) $\lim_{x \rightarrow 1^+} \frac{1}{x-1}$.

(b) $\lim_{x \rightarrow 1^-} \frac{1}{x-1}$.

(c) $\lim_{x \rightarrow 0} \frac{1}{x^2}$.

Example 7 Find the following limits.

$$(a) \lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2-4}.$$

$$(b) \lim_{x \rightarrow 2} \frac{x-2}{x^2-4}.$$

$$(c) \lim_{x \rightarrow 2^+} \frac{x-3}{x^2-4}.$$

$$(d) \lim_{x \rightarrow 2^-} \frac{x-3}{x^2-4}.$$

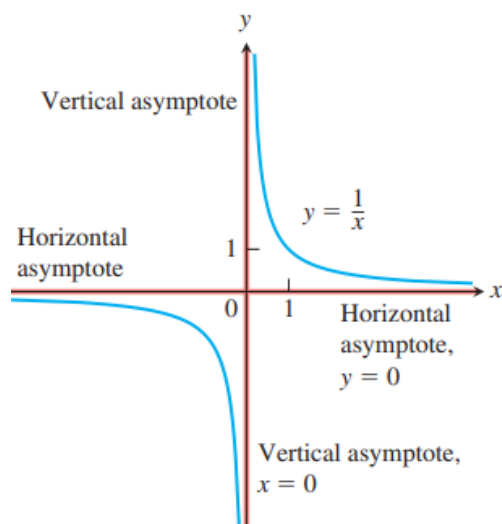
$$(e) \lim_{x \rightarrow 2} \frac{x-3}{x^2-4}.$$

$$(f) \lim_{x \rightarrow 2} \frac{2-x}{(x-2)^2}.$$

⑥ Vertical Asymptotes

DEFINITION A line $x = a$ is a vertical asymptote of the graph of a function $y = f(x)$ if either $\lim_{x \rightarrow a^+} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^-} f(x) = \pm\infty$.

For example, the graph of $y = \frac{1}{x}$ has a vertical asymptote $x = 0$.



◆ The key to find the vertical asymptote is let the bottom of a fraction function = 0.

Example 8 Find the horizontal and vertical asymptotes of the curve

$$y = \frac{x+3}{x+2}.$$

Example 9 Find the horizontal and vertical asymptotes of the graph of

$$f(x) = -\frac{8}{x^2 - 4}.$$